Lecture 01 — Teacher Reading Script

Private copy. Keep this PDF on your laptop or a second screen. Project only `Lecture_01_Trigonometry_Ratios_Identities.pptx`.

Class: Class 11 Physics · CBSE + NEET · Excellence Programme

Chapter: Basic Maths and Calculus · Trigonometry

Date: 1 September 2026

Deck: 65 slides · one 2-hour sitting (you manage the clock)

Break: Slide 25

How to use this script

- Page 1 of this PDF is this guide. After that, PDF page = deck slide + 1 (Slide 10 is PDF page 11). Each of those pages is one deck slide. The heading is the deck title.
- Say is the spoken line. Read it in full language, not in short form.
- Write is what you add on the empty board. The slide already shows the printed part.
- Wait means do not click next. Let them try.
- Private is the answer or the trap. Never read a Private block while the question slide is up.
- Speak cosec, never “csc”.
- On a Question slide, do not tick an option and do not flash the solution.
- Tonight slides: assign the work. Do not solve them in class. The answers are in Private only so you are ready after class.

Before they sit

- Stylus ready. Empty board is for you.
- Presenter View on, this PDF on the side.
- Daily-test reminder at the end, not at the start.

Slide 1 — Cover

On the board: EXCELLENCE · Class 11 · 2026–27 · Physics | Lecture - 01 · Basic Maths and Calculus · Trigonometry · By · date.

Say

“Good afternoon. This is Lecture 01 of Class 11 Physics. The subject is Physics. The chapter is Basic Maths and Calculus. Today’s topic inside that chapter is trigonometry.”

“You will write every formula and every question first. I will not rush the answer onto the board.”

Write

Nothing extra. Let the cover sit.

Transition

“Next slide is the list of topics. Then we start from the triangle.”

Slide 2 — Topics to be covered

On the board: eight numbered topics.

Say

“These are the topics for this lecture. We will do all of them today.”

Read the list once, slowly:

“One: trigonometric ratios — sine, cosine, tangent, cosecant, secant, cotangent.

Two: identities from P squared plus B squared equals H squared.

Three: finding unknown sides from a given ratio.

Four: the range of sine and cosine.

Five: algebra with ratios, and similar triangles.

Six: height and distance.

Seven: the standard-angle table from 0° to 180°, and 37° and 53°.

Eight: radian measure, arc equals R theta, and degree–radian conversion.”

“You do not need to photograph this. You will write each piece as we reach it.”

Write

Nothing. Move on.

Transition

“Open a fresh page. Title it Trigonometry. We start by naming the sides.”

Slide 3 — Name the sides from θ

On the board: right triangle, P orange, B blue, H green, 90° at the left, θ at the base.

Say

“Every trig ratio is named from one angle. That angle is theta. Do not name the sides from how the figure sits on the page. Name them from theta.”

“The side opposite theta is the perpendicular. We write P.

The side next to theta, that is not the hypotenuse, is the base. We write B.

The longest side, opposite the right angle, is the hypotenuse. We write H.”

“If I rotate this triangle, P is still opposite theta. That is the trap. Students lose marks because they call the bottom side the base even when theta has moved.”

Write

Point at each side as you say P, B, H. If needed, redraw the same triangle standing on a different side and relabel from theta.

Ask

“If I put theta at the top angle instead, which side becomes P?”

Wait. They should point to the side opposite that new theta.

Transition

“Now the six ratios. First pair: sine and cosecant.”

Slide 4 — sin θ and cosec θ

On the board: sin θ = P/H, cosec θ = H/P, product = 1.

Say

“Sine of theta is perpendicular over hypotenuse. Opposite over hypotenuse. Write it as sin theta equals P over H.”

“Cosecant is the reciprocal. Hypotenuse over perpendicular. If sine is 3/5, then cosecant is 5/3.”

“Their product is 1. Sine times cosecant equals 1. That is always true whenever sine is defined and not zero.”

“These are ratios. They have no unit. Never write sine theta equals 0.6 metre.”

Write

$\sin \theta = P/H$

$\operatorname{cosec} \theta = H/P$

$\sin \theta \cdot \operatorname{cosec} \theta = 1$

One example: $\sin \theta = 3/5 \rightarrow \operatorname{cosec} \theta = 5/3$

Transition

“Same pattern for cosine and secant.”

Slide 5 — cos θ and sec θ

On the board: cos θ = B/H, sec θ = H/B, product = 1.

Say

“Cosine of theta is base over hypotenuse. Adjacent over hypotenuse.”

“Secant is the reciprocal. If cosine is 4/5, secant is 5/4.”

“Cosine times secant equals 1. Again, no unit. Never write cosine theta equals 0.8 metre.”

Write

`cos θ = B/H`

`sec θ = H/B`

`cos θ · sec θ = 1`

Transition

“Last pair: tangent and cotangent.”

Slide 6 — tan θ and cot θ

On the board: tan θ = P/B, cot θ = B/P, product = 1.

Say

“Tangent of theta is perpendicular over base. Opposite over adjacent. This is the ratio we use most in height and distance.”

“Cotangent is the reciprocal. Also remember: tangent is sine over cosine.”

“Three products to lock: sine times cosecant is 1, cosine times secant is 1, tangent times cotangent is 1.”

Write

`tan θ = P/B`

`cot θ = B/P`

`tan θ = sin θ / cos θ`

and the three product lines.

Transition

“Next we get the identities. We do not memorise them as three separate miracles. All three come from Pythagoras.”

Slide 7 — Divide Pythagoras by H^2

On the board: $P^2 + B^2 = H^2$ divided by H^2 → $\sin^2\theta + \cos^2\theta = 1$.

Say

“Start from the right triangle. Perpendicular squared plus base squared equals hypotenuse squared.”

“Divide every term by H squared. Then P over H is sine, B over H is cosine, H over H is 1.”

“So sine squared plus cosine squared equals 1. This is the identity you keep even if you forget the other two.”

“From this you can rearrange: cosine is the square root of 1 minus sine squared, and sine is the square root of 1 minus cosine squared. For an acute angle we take the positive root.”

Write

The three lines of the division, then box $\sin^2\theta + \cos^2\theta = 1$.

Transition

“Same starting line. Now divide by B squared.”

Slide 8 — Divide Pythagoras by B^2

On the board: $1 + \tan^2\theta = \sec^2\theta$.

Say

“Same Pythagoras. Divide every term by B squared.”

“P over B is tangent. B over B is 1. H over B is secant. So 1 plus tangent squared equals secant squared.”

Write

The division, then box $1 + \tan^2\theta = \sec^2\theta$.

Transition

“Once more. Divide by P squared.”

Slide 9 — Divide Pythagoras by P^2

On the board: $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.

Say

“Divide by P squared. B over P is cotangent. H over P is cosecant. So 1 plus cotangent squared equals cosecant squared.”

“If your mind can hold only one identity in a paper, hold sine squared plus cosine squared equals 1. The other two are the same Pythagoras with a different division.”

Write

Box `1 + cot²θ = cosec²θ`.

Transition

“First question. You write. I wait.”

Slide 10 — Question: Find x

On the board: P = 1, B = 2, H = x. Find x, sin θ, cos θ, tan θ.

Say

“In the right triangle shown, the perpendicular is 1 and the base is 2. The hypotenuse is x. Find the value of x, sine theta, cosine theta and tangent theta.”

“Copy the triangle. Do not wait for me to write the answer.”

Wait

Do not click. Walk. If someone is stuck: “Which theorem gives you the hypotenuse?”

Private

$x = \sqrt{5}$, $\sin\theta = 1/\sqrt{5}$, $\cos\theta = 2/\sqrt{5}$, $\tan\theta = 1/2$.

Transition

Only after most notebooks have an attempt.

Slide 11 — Solution: First find H

Say

“Pythagoras. 1 squared plus 2 squared equals x squared. 1 plus 4 equals x squared. x equals square root of 5. We take the positive root because a length is positive.”

Write

$1^2 + 2^2 = x^2$

$x = \sqrt{5}$

Transition

“Now the three ratios.”

Slide 12 — Solution: Now sin θ, cos θ, tan θ

Say

“Sine is P over H, so 1 over root 5. Cosine is B over H, so 2 over root 5. Tangent is P over B, so 1 over 2.”

“Check: sine over cosine should be tangent. 1 over root 5, divided by 2 over root 5, is 1 over 2. Same. That check takes ten seconds and catches a wrong ratio.”

Write

The three ratio lines. Circle tan θ = 1/2.

Slide 13 — Question: If tan θ = 2

On the board: If tan θ = 2, find P, B, H, sin θ and cos θ. θ acute.

Say

“If tangent theta equals 2, then find the perpendicular, the base and the hypotenuse. Also find sine theta and cosine theta. Take theta acute.”

“You are not given a number for the base. So the sides will have a scale k. That is correct. The ratios will cancel k.”

Wait

Do not hint k yet unless the room is frozen. Then: “Tangent is 2 over 1. Let the 1 be k.”

Private

P = 2k, B = k, H = √5 k, sin θ = 2/√5, cos θ = 1/√5.

Slide 14 — Solution: tan θ = 2 → find P, B, H

Say

“Tangent is P over B equals 2 over 1. Let B equal k. Then P equals 2k. Hypotenuse squared is 4k squared plus k squared, which is 5k squared. Hypotenuse is root 5 times k. Positive, because theta is acute.”

Write

`P = 2k, B = k, H = √5 k`

Slide 15 — Solution: Now sin θ and cos θ

Say

“Sine is 2k over root 5 k. The k cancels. Sine is 2 over root 5. Cosine is k over root 5 k, so 1 over root 5.”

“A ratio only fixes the shape. The scale always cancels in sine, cosine and tangent. That is why a question can give you only tan theta equals 2.”

Write

`sin θ = 2/√5`

`cos θ = 1/√5`

Slide 16 — Question: Range of sin θ

On the board: If sin θ = 2, find cos θ and tan θ.

Say

“If sine theta equals 2, then find the value of cosine theta and tangent theta.”

Read it as a normal question. Do not smile. Do not say trap. Do not say impossible.

Wait

Let them try the identity. Many will write cosine squared equals 1 minus 4.

Private

Impossible. −1 ≤ sin θ ≤ 1. Do not click until they have tried.

Slide 17 — Solution: Method 1 — geometry

Say

“Before you open the identity, ask: can sine equal 2?”

“Sine is P over H. H is the longest side, so H is greater than P. A longer side over a shorter side cannot be 2. P over H is at most 1.”

“There is no such triangle. Sine theta equals 2 is impossible. Cosine and tangent are not defined for this given.”

Write

$H > P \Rightarrow P/H \leq 1$

$\sin\theta = 2$ is impossible

Slide 18 — Solution: Method 2 — identity

Say

“If someone still uses the identity: sine squared plus cosine squared equals 1, put 2, get 4 plus cosine squared equals 1, cosine squared equals minus 3. That is not a real number. Same conclusion.”

“Lock the range. Sine lies between minus 1 and 1, including the ends. Cosine the same. Tangent has no finite range. Tangent 90° is not defined.”

Write

$-1 \leq \sin \theta \leq 1$

$-1 \leq \cos \theta \leq 1$

$\tan 90^\circ$ not defined

Transition

“Next question is algebra. You will turn a sine–cosine fraction into tangent.”

Slide 19 — Question: Find tan θ from a sin–cos fraction

On the board: (sin θ + cos θ)/(sin θ − cos θ) = 11/5. Find tan θ.

Say

“If sine plus cosine, over sine minus cosine, equals 11 over 5, then find the value of tangent theta.”

“Do not find sine and cosine separately. Aim for sine over cosine.”

Wait

Private

tan θ = 8/3.

Slide 20 — Solution: Cross multiply, then expand

Say

“Cross multiply. 5 times sine plus cosine equals 11 times sine minus cosine.”

“Open the brackets. 5 sine plus 5 cosine equals 11 sine minus 11 cosine.”

“Next slide we collect. Do not jump.”

Write

$$5(\sin\theta + \cos\theta) = 11(\sin\theta - \cos\theta)$$

$$5\sin\theta + 5\cos\theta = 11\sin\theta - 11\cos\theta$$

Slide 21 — Solution: Collect, then divide

Say

“Bring cosine terms to the left, sine terms to the right. 5 cosine plus 11 cosine is 16 cosine. 11 sine minus 5 sine is 6 sine.”

“16 cosine equals 6 sine. Divide both sides by cosine, and swap: sine over cosine equals 16 over 6, which is 8 over 3.”

“Tangent theta is 8 over 3.”

Write

`16 cos θ = 6 sin θ`

`tan θ = 8/3`

Slide 22 — Question: Nested right triangles

On the board: large triangle opposite 4, adjacent 3. Small slice opposite 2, adjacent y. Middle slice opposite x, adjacent 2. Four options. None ticked.

Say

“A large right triangle has opposite side 4 and adjacent side 3. Nested similar slices share the same angle theta. In the small slice the opposite is 2 and the adjacent is y. In the middle slice the opposite is x and the adjacent is 2. Find the values of x and y.”

“I will draw the nested figure on the board. You copy it. Same theta means same tangent. Do not mix a height from one slice with a base from another.”

Write

Sketch: one large right triangle, two similar inner slices sharing theta. Label 4, 3 on the large; 2 and y on the small; x and 2 on the middle.

Wait

Options stay unticked.

Private

Option 3: $x = 8/3$, $y = 3/2$.

Slide 23 — Solution: Find y from tan θ = 4/3

Say

“Large triangle: tangent theta is 4 over 3. Small slice: tangent is 2 over y. Same theta, so 4 over 3 equals 2 over y. Cross multiply: 4y equals 6, y equals 3 over 2.”

Write

`4/3 = 2/y`

`y = 3/2`

Slide 24 — Solution: Find x from tan θ = 4/3

Say

“Middle slice: opposite x, adjacent 2. Tangent is x over 2. Still 4 over 3. So 4 over 3 equals x over 2. 3x equals 8. x equals 8 over 3.”

“The trap is mixing triangles: 4 with 2, or x with 3. Each ratio from the same triangle.”

Write

`4/3 = x/2`

`x = 8/3`

Circle option 3 only now, if you wish, on your board, not as a pre-tick on the ask slide.

Transition

“That is the first sitting. Break. After the break: height and distance, the table, 37 and 53, then radian.”

Slide 25 — BREAK

Say

“Break. Stretch. Do not pack bags. We continue with towers, then the 0 to 180 table.”

Write

Nothing. Leave the slide up.

Slide 26 — Question: Observer between two towers — elevation

On the board: two towers d apart, observer x from the nearer, elevations θ and ϕ. Four options.

Say

“From a point on level ground between two vertical towers that are d metres apart, the angles of elevation to the tops of the nearer and farther towers are theta and phi respectively. The observer is x metres from the nearer tower. What is the difference in heights of the towers?”

“Draw first. Two vertical lines. A point between them. x to the nearer, d minus x to the farther. Elevation is the angle up from the ground.”

Write

Two towers, observer between them, θ and ϕ marked from the ground.

Wait

Private

Option 2: $x \tan\theta - (d - x)\tan\phi$. (This is nearer minus farther. If a paper asks farther minus nearer, flip the sign.)

Slide 27 — Solution: Nearer tower

Say

“Nearer tower. Opposite is height h1. Adjacent is x. Tangent theta equals h1 over x. So h1 equals x tan theta.”

Write

`tan θ = h■ / x`

`h■ = x tan θ`

Slide 28 — Solution: Farther tower, then the difference

Say

“Farther tower. Adjacent is now d minus x. Elevation is phi. So h2 equals (d minus x) tan phi.”

“Difference, nearer minus farther: x tan theta minus (d minus x) tan phi. That is option 2.”

“If the question asks for the farther minus the nearer, flip the sign. Always sketch before you write tan.”

Write

$h\blacksquare = (d - x) \tan \phi$

$h\blacksquare - h\blacksquare = x \tan \theta - (d - x) \tan \phi$

Slide 29 — Question: From the top of one tower — depression

On the board: two towers, distance D, depression θ from the top of h1 to the top of h2. h1 > h2. Four options.

Say

“Two towers are separated by a horizontal distance D. From the top of the first tower, height h1, the angle of depression to the top of the second tower, height h2, is theta. Which relation correctly gives D in terms of h1, h2 and theta? Take h1 greater than h2.”

“Depression looks down. The opposite side is the drop in height, h1 minus h2, never the sum.”

Write

Two towers of different heights. Horizontal from the top of the taller to a point above the shorter. The drop h1 − h2 and the run D.

Wait

Private

Option 2: $D = (h_1 - h_2) / \tan \theta$.

Slide 30 — Solution: Write tan θ first

Say

“From the top of the first tower, look down to the top of the second. Opposite is the height drop h1 minus h2. Adjacent is D. Tangent theta equals (h1 minus h2) over D.”

Write

```
tan θ = (h■ – h■) / D
```

Slide 31 — Solution: Solve for D

Say

“Rearrange. D equals (h1 minus h2) over tan theta.”

“Wrong options add the heights, or multiply by tan theta. Adding heights would be as if one tower stood on top of the other. That is not this figure.”

Write

```
D = (h■ – h■) / tan θ
```

Slide 32 — Table: sin, cos, tan from 0° to 90°

On the board: 0°, 30°, 45°, 60°, 90°.

Say

“We fill this column by column, not row by row. That way the pattern of sine going up and cosine going down is visible.”

Walk 0°: “Sine 0 is 0, cosine 0 is 1, tangent 0 is 0.”

Walk 30°: “Sine 1/2, cosine root 3 over 2, tangent 1 over root 3.”

Walk 45°: “Sine and cosine both 1 over root 2. Tangent 1.”

Walk 60°: “Sine and cosine swap with 30°. Tangent root 3.”

Walk 90°: “Sine 1, cosine 0, tangent is not defined. Write undef. Do not write infinity as if it were a number you can substitute.”

Write

Fill with them if the table is blank on your board copy. On the slide the values are already there — still speak each cell.

Transition

“After 90°, cosine becomes negative. Next table.”

Slide 33 — Table: sin, cos, tan from 90° to 180°

On the board: 90° to 180°, including negatives. Range lock.

Say

“Second quadrant. Sine stays positive. Cosine is negative. Tangent is negative.”

“120° is the brother of 60°. Sine is still root 3 over 2. Cosine is minus 1/2. Tangent is minus root 3.”

“135° is the brother of 45°. Sine 1 over root 2, cosine minus 1 over root 2, tangent minus 1.”

“150° is the brother of 30°. Sine 1/2, cosine minus root 3 over 2.”

“180°: sine 0, cosine minus 1, tangent 0.”

“Range lock again: sine and cosine never leave minus 1 to 1.”

Write

Point at the sign of cosine. Underline tan 90° undef.

Slide 34 — Question: Read these from the table

On the board: (i) to (v) — sin 120°, cos 150°, tan 135°, sin 150°, cos 120°.

Say

“From the 0° to 180° table, find the value of each of the following.”

“Copy the five. Look at the table. Do not invent a calculator.”

Wait

Private

sin 120° = √3/2, cos 150° = −√3/2, tan 135° = −1, sin 150° = 1/2, cos 120° = −1/2.

Slide 35 — Solution: From the 0°–180° table

Say

Read each answer once. Stress the minus signs on cosine 150, tangent 135, cosine 120.

“If you missed a minus, you used the 0 to 90 table for an angle past 90. The brother angle gives the size. The quadrant gives the sign.”

Slide 36 — Question: 3–4–5 triangle — 37° and 53°

On the board: 3–4–5 triangle. Opposite 4 is the vertical side on this figure. Adjacent 3 is the base.

Say

“In a 3–4–5 right triangle, find sine, cosine and tangent of 37°, and sine, cosine and tangent of 53°.”

“The small acute angle is 37°. It is opposite the side 3. The large acute angle is 53°. It is opposite the side 4. 37 plus 53 is 90, so they are complementary.”

“On this figure the labelled theta at the base is opposite 4, so that theta is 53°. Do not blindly call every theta 37°.”

Wait

Private

sin 37° = 3/5, cos 37° = 4/5, tan 37° = 3/4.

sin 53° = 4/5, cos 53° = 3/5, tan 53° = 4/3.

Slide 37 — Solution: 3–4–5 triangle — 37° and 53°

Say

“Opposite 3, hypotenuse 5: sine 37 is 3/5. Adjacent to 37 is 4: cosine 37 is 4/5. Tangent 37 is 3/4.”

“Opposite 4, hypotenuse 5: sine 53 is 4/5. Adjacent to 53 is 3: cosine 53 is 3/5. Tangent 53 is 4/3.”

“Tangent 37 is less than tangent 53 because 3/4 is less than 4/3. Also sine 53 times cosecant 53 is 1 — that is tonight’s reminder, not a new identity.”

Write

The two rows of ratios. Do not write the tan 37 comparison across the triangle — it lives on the right, under the ratios.

Slide 38 — Question: From 0° to 90°

On the board: If the angle increases from 0° to 90°, what happens to sin θ and cos θ? Four options.

Say

“If the angle increases from 0° to 90°, then what happens to the value of sine theta and the value of cosine theta?”

“Look back at the table we just filled. Do not guess.”

Wait

Private

Option 1: sin θ increases, cos θ decreases.

Slide 39 — Solution: From 0° to 90°

Say

“Sine goes 0, 1/2, 1 over root 2, root 3 over 2, 1. It increases. Cosine goes 1, root 3 over 2, 1 over root 2, 1/2, 0. It decreases. Tangent increases, and tangent 90 is not defined.”

Write

`0° → 90°: sin ↑ cos ↓ tan ↑`

Slide 40 — Question: Identity

On the board: Find the value of $\sin^2 40° + \cos^2 40°$.

Say

“Find the value of sine squared 40° plus cosine squared 40°.”

Do not let them reach for a calculator. If someone starts computing 40°, stop them with: “Is the angle the same in both terms?”

Wait

Private

1. Same angle. Identity. Do not compute 40°.

Slide 41 — Solution: $\sin^2 40° + \cos^2 40°$

Say

“True for every theta. Here theta is 40°. Sine squared plus cosine squared equals 1. Do not compute 40°.”

Write

$\sin^2 40° + \cos^2 40° = 1$

Slide 42 — Arc length

On the board: Arc = Rθ, θ in radian.

Say

“Practical unit of angle is degree. SI unit of angle is radian. When we write arc length equals R times theta, theta must be in radian. Never put 90° into this formula.”

“Positive theta is anti-clockwise. Negative theta is clockwise. For distance along the path we will use the positive radian measure of the turn.”

Write

`Arc = Rθ` with θ in radian.

`degree = practical, radian = SI`

Slide 43 — The one line: 180° = π rad

Say

“Lock this one line. 180° equals pi radian. Wherever you see pi radian, put 180°. Every conversion today starts from this line. Do not memorise a dozen separate facts.”

Write

Box `180° = π rad`.

Slide 44 — Degree → radian

Say

“Divide the lock-line by 180. One degree equals pi over 180 radian. So to convert a degree measure into radian, multiply by pi over 180.”

Write

`1° = π/180 rad`

`θ° in radian = θ × (π/180)`

Slide 45 — Radian → degree

Say

“Divide the lock-line by pi. One radian equals 180 over pi degree. To convert radian into degree, multiply by 180 over pi. Same as: replace pi radian by 180°.”

Write

`1 rad = 180°/π`

`θ rad in degree = θ × (180/π)`

Slide 46 — Question: Arc on a circle

On the board: object on circumference, radius R, turns 90°. Find the distance.

Say

“An object moves on the circumference of a circle of radius R. If the object turns through 90°, find the total distance moved by the object.”

Do not hint that 90° cannot go into R theta. Let the trap appear in their notebooks.

Wait

Private

Distance = πR/2. Wrong: R × 90°.

Slide 47 — Solution: Object turns 90° on a circle of radius R

Say

“Distance is the path, which is the arc, which is R theta. Theta must be radian. 90° is pi over 2 radian. Distance is R times pi over 2, which is pi R over 2.”

“If anyone wrote R times 90, cross it out. Degree and metre do not multiply to a length in this formula.”

Write

`90° = π/2 rad`

`s = Rθ = πR/2`

Slide 48 — Question: First conversion

On the board: Convert π/3 radian into degree.

Say

“Convert pi by 3 radian into degree.”

Wait

Private

60°.

Slide 49 — Solution: π/3 rad

Say

“Wherever you see pi radian, put 180°. Pi over 3 becomes 180 over 3, which is 60°.”

Write

```
π/3 rad = 180°/3 = 60°
```

Slide 50 — Question: Convert these now

On the board: (i) π/2 (ii) π/4 (iii) π/6 (iv) 45° (v) 90° (vi) 60°.

Say

“Convert the following into degree or radian, as required. These six are classwork. The longer list after this is tonight. Do not mix the two.”

Wait

Live board. Do not open the solution until most have all six.

Private

π/2 = 90°, π/4 = 45°, π/6 = 30°, 45° = π/4, 90° = π/2, 60° = π/3.

Slide 51 — Solution: The six conversions from the board

Say

Read the six answers. “Same lock-line every time. Pi over 2 is 90. Pi over 4 is 45. Pi over 6 is 30. The three degree-to-radian lines are just the reverse.”

“Tonight’s list is longer. It stays unanswered.”

Slide 52 — Tonight: Find β

On the board: triangle 40° and 70°, side next to 70° extended, find exterior β.

Say

“A triangle has two interior angles 40° and 70°. The side next to the 70° angle is extended. If beta is the exterior angle formed at that vertex, find the value of beta.”

“This is tonight. Copy the question. Draw the figure at home. I will not solve it now.”

Private

Third interior angle = 70°. Exterior β = 110°. (Exterior = sum of remote interiors = 40° + 70°, or 180° − 70° interior at that vertex.)

Slide 53 — Tonight: Convert radian into degree

On the board: (i) to (ix) radian measures.

Say

“Convert the following radian measures into degree. Nine items. Tonight. Same method as pi over 3.”

Private

5π/6 = 150°, 2π/3 = 120°, 4π/3 = 240°, 3π/5 = 108°, 5π = 900°, 2π/5 = 72°, 7π/12 = 105°, 5π/2 = 450°, 3π = 540°.

Slide 54 — Tonight: Convert degree into radian

On the board: 120°, 30°, 150°, 270°, 135°, π.

Say

“Convert the following into radian. The last item is already pi. Pi is a radian measure. Do not convert pi into something else unless the question asks for a number times 1 radian — and here it is already radian.”

Private

120° = 2π/3, 30° = π/6, 150° = 5π/6, 270° = 3π/2, 135° = 3π/4, π is already radian.

Slide 55 — Tonight: Product of a ratio and its reciprocal

Say

“Find the value of sine 53° times cosecant 53°. Tonight. You already saw the identity sine times cosecant equals 1.”

Private

1.

Slide 56 — Tonight: Product of three values

Say

“Find the value of sine 180° times cosine 135° times tangent 120°. Tonight. Use the table.”

Private

sin 180° = 0, so the product is 0.

Slide 57 — Tonight: Shadow of a pole

On the board: pole h, shadow l, Sun at θ. Four options. None ticked.

Say

“A vertical pole of height h casts a shadow of length l when the Sun is at an angle theta above the horizon. Which of the following expresses the shadow length l in terms of h and theta?”

“Tonight. Draw the right triangle: height h, shadow l, angle theta at the Sun end of the shadow.”

Private

Option 2: l = h / tan θ (same as h cot θ).

Slide 58 — Tonight: Wire from the top of a pole

Say

“A pole of height h has a wire attached from the top to the ground at an angle theta. If the wire breaks and falls to the ground without changing length, find the horizontal distance x from the base of the pole where the wire touches the ground.”

“Tonight. The wire length is the hypotenuse. After it falls, that same length lies on the ground. x is not that full length — read the question again at home.”

Private

Wire length = h / sin θ. After it falls, the free end is that far from the top’s ground-point; distance from the base of the pole is x = h cot θ. Option 1.

Slide 59 — Tonight: Shadow on sloping ground

Say

“A vertical pole of height h = 10 m stands on ground that slopes upwards at a constant angle alpha = 30° with the horizontal. If the Sun’s angle of elevation above the horizontal is theta = 60°, what is the length of the shadow cast by the pole on the sloping ground?”

“Tonight. Four options. Draw: slope 30°, sun 60° to the horizontal, pole 10 m. Do not pick 10 m or 10 root 3 by guessing.”

Private

Option 4: 10 / (√3 − 1) m.

Slide 60 — Tonight: After class

Say

“Daily test before the next lecture. Complete it tonight after you finish the board questions.”

Write

Nothing.

Slide 61 — Today: P, B, H

On the board: the six ratios.

Say

“Close the notebooks for thirty seconds. From theta: P opposite, B adjacent, H hypotenuse. Sine P over H, cosine B over H, tangent P over B, and the three reciprocals.”

If you want a chorus, ask three students for sine, cosine, tangent.

Slide 62 — Today: Identities

Say

“Three identities, one Pythagoras. Sine squared plus cosine squared equals 1. 1 plus tangent squared equals secant squared. 1 plus cotangent squared equals cosecant squared. If you keep only one line, keep the first.”

Slide 63 — Today: Range and table

Say

“Sine and cosine live between minus 1 and 1. The 0 to 180 table must be in the hand, not only in the phone. Tangent 90 is not defined.”

Slide 64 — Today: Radian

Say

“Arc equals R theta with theta in radian. Pi radian means 180°. That is the whole conversion engine.”

Slide 65 — Today: After class

Say

“Daily test tonight. Remaining conversions and the PYQs tonight. Do not leave the 3–4–5 ratios unwritten. Next lecture we continue from here.”

“Pack. See you next class.”

After they leave

- Do not post tonight answers in the group.
- Tomorrow’s opening: one wrong daily-test item, then new topic.
- If someone asks β or the sloping-ground PYQ after class, use the Private blocks above. Sketch first, then the line.